

ABDULLAH MUSHTAQ

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RESEARCH INTERESTS

Natural Language Processing · Large Language Models · Multi-Agent Systems · Alignment and Hallucination · AI4Science

EDUCATION

Tulane University

Ph.D. in Computer Science | Advisor: [Aron Culotta](#)

New Orleans, LA, USA

Aug. 2026 – Present

Information Technology University

Bachelor of Science in Computer Science

Dean's Honor List | Rank: Top 5% | Specialization GPA: 3.83/4.0 | Merit Scholarship 2020-2024

Lahore, Pakistan

2020 – 2024

The Educators College

Intermediate Education, Intermediate in Computer Science | Grade: A+ | Topper in Computer Science

Chichawatni, Pakistan

2018 – 2020

Dawn High School

Secondary Education | Grade: A+ | District Topper in Computer Science & Physics; School Top Scorer in Urdu

Chichawatni, Pakistan

2016 – 2018

PUBLICATIONS

- [1] Does the Language Carry the Worldview? A Multilingual Benchmark for Culturally Grounded Therapeutic Reasoning in Large Language Models. *PNAS Nexus* (Under Review).
- [2] Elmahjub, Ezieddin, Qadir, Junaid, **Mushtaq, Abdullah**, Naeem, Rafay, Ghaznavi, Ibrahim, & Iqbal, Waleed. (2026). IslamicLegalBench: Evaluating LLMs knowledge and reasoning of Islamic law across 1,200 years of Islamic pluralist legal traditions. *Artificial Intelligence and Law*. Springer. (Open Access) DOI: [10.1007/s10506-026-09535-4](https://doi.org/10.1007/s10506-026-09535-4)
- [3] **Mushtaq, Abdullah**, Taj, Imran, Naeem, Rafay, Ghaznavi, Ibrahim, & Qadir, Junaid. (2026). WorldView-Bench: A benchmark for evaluating global cultural perspectives in large language models. *Journal of Artificial Intelligence Research*, Vol. 85, Page 28. (Acceptance Rate: 12.6%, Q1, H-index: 143) DOI: [10.1613/jair.1.19001](https://doi.org/10.1613/jair.1.19001)
- [4] **Mushtaq, Abdullah**, Naeem, Rafay, Elmahjub, Ezieddin, Ghaznavi, Ibrahim, Al-Maliki, Shawqi Mohammed, Abdallah, Mohamed, Al-Fuqaha, Ala, & Qadir, Junaid. (2025). Can LLMs Write Faithfully? An Agent-Based Evaluation of LLM-generated Islamic Content. In *5th Muslims in ML Workshop co-located with NeurIPS 2025*. [arXiv:2510.24438](https://arxiv.org/abs/2510.24438)
- [5] **Mushtaq, Abdullah**, Naeem, Rafay, Ghaznavi, Ibrahim, Taj, Imran, Hashmi, Imran, & Qadir, Junaid. (2025, April). Harnessing multi-agent LLMs for complex engineering problem-solving: A framework for senior design projects. In *2025 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1–10). IEEE. [arXiv:2501.01205](https://arxiv.org/abs/2501.01205)
- [6] **Mushtaq, Abdullah**, Naeem, Muhammad Rafay, Taj, Muhammad Imran, Ghaznavi, Ibrahim, & Qadir, Junaid. (2025). Toward inclusive educational AI: Auditing and alignment of frontier LLMs for cultural biases through a multiplexity lens. In *2025 IEEE Global Engineering Education Conference (EDUCON)* (pp. 1–10). IEEE. [arXiv:2501.03259](https://arxiv.org/abs/2501.03259)
- [7] **Mushtaq, Abdullah**, Naeem, Muhammad Rafay, Ghaznavi, Ibrahim, Abd-alrazaq, Alaa, Tabassum, Aliya, & Qadir, Junaid. (2025). Can Agents Judge Systematic Reviews Like Humans? Evaluating SLRs with LLM-based Multi-Agent System. (In-progress) [arXiv:2509.17240](https://arxiv.org/abs/2509.17240)

EXPERIENCE & RESEARCH WORK

Tulane University, TulaneAI Group — New Orleans, LA, USA

Graduate Research Assistant — Natural Language Processing

Aug. 2026 – Present

Supervisor & PI: [Aron Culotta](#) ([Tulane University](#))

Project: Socio-linguistic modeling to understand the long-term dynamics of news engagement in online media

NSF-HCC

- Working at the intersection of natural language processing, large language models, and causal inference to study how people change their political views and approach social interactions on platforms like Reddit.
- Collected ~30k public comments and built the preprocessing pipeline, including PII scrubbing and filtering.
- Built an LLM annotation system with typed schemas that benchmarks state-of-the-art models against human-coded gold standards on agreement, precision, and recall.

Information Technology University (ITU) — Lahore, Pakistan

Research Assistant — Human-Computer Interaction & Natural Language Processing

Dec. 2025 – July 2026

Supervisors: [Ibrahim Ghaznavi](#) (ITU), [Junaid Qadir](#) (Qatar University)

Project: Does the Language Carry the Worldview? Multilingual Therapeutic Reasoning in LLMs *PNAS Nexus* (under review)

- Built a multilingual benchmark of first-person therapeutic scenarios across clinically important domains, rendered with pragmatic rather than literal equivalence, to test whether LLM therapeutic reasoning stays consistent when only the language of the request changes.

- Designed a six-dimension evaluation framework with reproducible scoring rubrics, informed by Traditional Islamically Integrated Psychotherapy; every response is scored by multiple LLM judges from disjoint model families. Introduced the Worldview Shift Index to measure language dependence without ranking worldviews.
- Found systematic language-driven shifts in how users and their distress are framed (Cliff's δ 0.29–0.50, $p_{\text{FDR}} < 0.001$) while relational warmth stayed stable, with high inter-judge agreement (0.957 vs. 0.953 quadratic-weighted).

Qatar University — Doha, Qatar

Visiting Research Assistant — LLM Reasoning & Agentic AI

Aug. 2025 – Nov. 2025

Supervisors: [Ezieddin Elmahjub \(Harvard & Qatar University\)](#), [Junaid Qadir \(Qatar University\)](#), [Ibrahim Ghaznavi \(ITU\)](#)

Project: IslamicLegalBench: Evaluating LLM Reasoning Across 1,200 Years of Islamic Law

AI & Law 2026 | [Paper](#) | [arXiv](#) | [Leaderboard](#)

- Constructed a multi-dimensional benchmark comprising 14 distinct task types, spanning basic factual recall to advanced jurisprudential reasoning, from Islamic legal texts across 1,200+ years and multiple schools of thought.
- Architected an end-to-end large-scale evaluation pipeline with reproducible protocols, semantic scoring, and statistical validation; rigorously evaluated frontier LLMs (GPT-5, Gemini 2.5, Claude 4, DeepSeek R1, Grok 4, Llama 4) under LLM-as-a-Judge, expert human annotation, and structured prompting strategies (Zero-shot, Few-shot, Chain-of-Thought).
- Served as **lead contributor** on the manuscript, published in *Artificial Intelligence and Law* (Springer, 2026), advancing the state of domain-specific legal reasoning in AI.

Project: SLR-GPT: Can LLM-Agents Judge Systematic Reviews Like Humans?

In-progress | [Paper](#)

Project Collaborators: [Alaa A. Abd-alrazaq \(Weill Cornell Medicine\)](#), [Aliya Tabassum \(Qatar University\)](#)

- Designed and built a Multi-Agent System of 27 specialized LLM-based agents organized into deliberative societies, integrating few-shot prompting, real-time tool use, MCP, Vision Language Models, and an ArXiv Toolkit into a unified multimodal agentic pipeline for comprehensive SLR analysis from a single PDF input.
- Engineered an interactive Flask-based interface enabling seamless agent-user interaction, structured paper analysis, and interpretable insight extraction at scale.
- Achieved **84%** agreement with domain experts across medicine, AI, and AR/VR systematic reviews, demonstrating strong reliability for AI-assisted scholarly review assessment.

Hamad Bin Khalifa University (HBKU) — Doha, Qatar (remote)

Research Assistant — Agentic AI & Large Language Models

May 2025 – Aug. 2025

Supervisors: [Mohamed Abdallah \(HBKU\)](#), [Ala Al-Fuqaha \(HBKU\)](#), [Shawqi M. Al-Maliki \(HBKU\)](#), [Junaid Qadir \(Qatar University\)](#)

Project: Can LLMs Write Faithfully? — Agent-Based Evaluation of LLM-generated Religious Content

NeurIPS 2025 (MusIML) | [Paper](#) | [Code](#)

- Developed a dual-agent evaluation framework (quantitative + qualitative) to assess LLM reliability in producing religious guidance, incorporating citation verification, multi-dimensional scoring, and structured interpretability schemas at scale.
- Benchmarked frontier and domain-specific LLMs on authentic religious essay prompts, identifying critical gaps in citation accuracy, religious consistency, and faith-sensitive content generation.

Zayed University — Abu Dhabi, UAE

Research Assistant — Large Language Models

July 2024 – July 2025

Supervisors: [Imran Hashmi \(Oxford\)](#), [Junaid Qadir \(Qatar University\)](#), [Imran Taj \(Zayed University\)](#)

Project: WorldView-Bench — Evaluating Global Cultural Perspectives in LLMs

JAIR 2026 | [Paper](#) | [Code](#)

- Introduced a benchmark of 175 questions across 7 categories for measuring cultural inclusivity in frontier LLMs; improved inclusivity metrics from **13%** to **94%** while reducing negative sentiment to **2.4%** via MAS-based multiplex approaches.

Project: MASC — Multi-Agent SDP Co-Pilot for Complex Engineering

IEEE EDUCON 2025 | [Paper](#) | [Code](#)

- Built a collective intelligence framework for automated assessment of engineering projects, achieving **89%** alignment with expert faculty evaluations; integrated advanced NLP metrics (Lexical Cohesion, Clause Density) and dynamic agent-based QnA.

Project: Cultural Bias Auditing — Alignment of Frontier LLMs

IEEE EDUCON 2025 | [arXiv](#)

- Developed methodologies for detecting and mitigating cultural biases in frontier LLMs (GPT-4, Claude 3.5, Llama 3.1 & 3.2, Mistral 7B), improving inclusivity rates from **3.25%** to **98%** via bias-aware prompt calibration, controlled generations, and Multi-Agent Systems.

CSaLT Lab, Lahore University of Management Sciences (LUMS) — Lahore, Pakistan

Undergraduate Research Assistant — Natural Language Processing

Aug. 2023 – Apr. 2024

Supervisors: [Agha Ali Raza \(LUMS & CMU\)](#), [Abbas Edalat \(Imperial College London\)](#)

Project: Urdu-language Psychotherapy Chatbot — Low-resource NLP for Mental Health Joint CSaLT-Imperial College London

- Fine-tuned and integrated state-of-the-art models (Whisper, M4T, BERT, MuRIL, GPT-2) on a custom Urdu dataset, incorporating sentiment analysis, embedding generation, and domain-specific language processing tailored for mental health support.
- Engineered the complete Python backend for information retrieval and collaborated on a Flutter-based front-end interface, delivering an accessible and seamless experience for native Urdu-speaking users.

Intelligent Machines Lab (IML), Information Technology University — Lahore, Pakistan

Research Intern — 3D Computer Vision

Aug. 2023 – Apr. 2024

Supervisor: [Mohsen Ali \(ITU & University of Florida\)](#)

Project: Real-time 3D Reconstruction & Neural Rendering — Joint IML-[ETRI](#) (South Korea) Research Project

- Developed a large-scale synthetic 4D dataset (~2TB) comprising 80+ high-fidelity indoor scenes in Unity3D using the Perception Toolkit, incorporating physics-based animations and multimodal outputs (RGB-D, segmentation, video).
- Benchmarked temporal reconstruction pipelines using SfM, COLMAP, NeRF, and 3D Gaussian Splatting with iterative refinement, focusing on neural radiance fields and temporal view synthesis for optimal reconstruction quality.

Information Technology University — Lahore, Pakistan

Teaching Assistant

Aug. 2023 – June 2024

- CS595 — Theory and Application of Virtual Reality (Spring 2024)
- CS104 — Application of Information and Communication Technologies (Fall 2023)
- CS152 — Object Oriented Programming (Spring 2022)

Virtuality Labs, Information Technology University — Lahore, Pakistan

Lab Lead - Natural Language Processing & Virtual Reality

2023 – 2024

Supervisors: [Ibrahim Ghaznavi](#) (ITU), [Abid Hussain](#) (Copenhagen Business School)

- Mentored **17 interns**, managed international research collaborations with institutions including Imperial College London, Copenhagen Business School, Toronto Metropolitan University, and Kristiania College Norway, and represented the lab in demonstrations to government delegations including the [Chief Minister of Punjab](#).

Project: First-ever VR Classroom Field Experiment in Pakistan

ECIS 2025 | [✕ Paper](#)

- Volunteered and Led a team of junior researchers in designing and conducting a large-scale field experiment comparing VR, Face-to-Face, and Zoom teaching conditions across 100+ first-year CS students.

ADDITIONAL PROJECTS

Task-Agnostic Continual Learning for Embodied Agents | *Python, PyTorch, DreamerV3, DeepMind Control, Meta-World*

- Proposed task-agnostic continual reinforcement learning grounded in a world model: a shared latent dynamics model trained jointly with the policy, whose structured prediction error serves as a self-supervised signal of task or regime change, combined with a compositional skill memory that retains and recombines learned behaviors without task IDs.
- Adapted a PyTorch DreamerV3 world model with a custom loader for non-stationary task streams, and trained it from 64×64 pixels on silently switching DeepMind Control tasks (walker-stand → cheetah-run → walker-stand, 25k steps each).
- Without task labels, the prior reconstruction error spiked **8×** and **95×** above baseline at the exact step of each switch, before the agent had a chance to fail, and a t-SNE of RSSM hidden states separated the tasks into distinct clusters.
- Reproduced LEGION (*Nature Machine Intelligence*) on Meta-World MT10 as a continual-RL baseline and began adapting it to train without task IDs.

Fine-tuned LLaMA 2(7B) for “Friends” Screenplay Scripts Generation | *Python, PyTorch, Transformers, LoRA*

- Fine-Tuned *LLaMA 2(7B)* LLM on 236 episodes to generate new scripts by episode name.
- Used QLoRA for efficient 4-bit quantization and LoRA-based parameter-efficient tuning. Leveraged GPU acceleration, adaptive learning rates, and gradient checkpointing for supervised fine-tuning.

Gaussian Optimization for Image Reconstruction | *Python, PyTorch, NumPy, CV2* | Lab Lead: [Usama Hassan](#) | [Code](#) 

- Co-developed a 2D Gaussian Splatting model for image reconstruction via GPU-accelerated backpropagation over mean and covariance tensors in PyTorch, inspired by the original [3D Gaussian Splatting](#) paper, and predating subsequent works applying the same technique to [compression](#) and [geometrically accurate radiance fields](#).

Photoshopped Images Detection | *Python, TensorFlow, Numpy, CV2, Matplotlib* | Advisor: [Mehwish Ghafoor](#)

- Fine-tuned Meta’s ConvNeXt model on a dataset of 2,000+ images for detecting Photoshop-based face manipulations (Face Aware Liquify), achieving **65%** accuracy after 10 training epochs and outperforming average human detection rates of 53.5%.
- Conducted comparative analysis against Transformer-based models following the methodology of a [ICCV 2019 paper](#).

AWARDS & HONORS

National Grassroot ICT Research Initiative (NGIRI) 2023–24 | *IGNITE*

2023 – 2024

- Awarded the NGIRI 2023–24 national research grant (a federal ICT R&D fund), receiving **100,000 PKR** for “*Harnessing Mixed Reality and Drones for Agriculture using Generative AI*”, selected among outstanding final-year projects across Pakistani universities.

Google Competitive Programming Competition | *Google Developer Student Club, ITU*

2022

- Secured **4th** position in a competitive programming among 150+ participants across the country.

Undergrad Ehsaas Scholarship | *Higher Education Commission*

2020 – 2024

- Secured a highly competitive, fully-funded scholarship for the bachelor’s based on history of academic excellence.

MEDIA & PRESS COVERAGE

- **National TV:** Research aired live on [ARY News](#) and [Samaa TV](#).
- **Print & Online:** Featured in [Daily Times](#) and [Academia Magazine](#).
- **Documentary:** Covered in an [official ITU YouTube documentary](#) titled “*VR Classes at ITU | 1st time in Pakistan*”.
- **Government:** Project demonstrated live to the [Chief Minister of Punjab](#).

EXTRACURRICULAR & VOLUNTEERING

Co-Founder & Head of Game Arts and Design

ITU Game Society | Information Technology University

Dec 2022 – Mar 2024

Lahore, Pakistan

- Co-founded the ITU Game Society and led all game arts and design operations, organizing game development workshops and eSports tournaments for 100s of students across the country.

TECHNICAL SKILLS

Languages: Python, C/C++ , C#

Libraries: PyTorch, TF, NumPy, Pandas, Sklearn, Transformers, OpenAI Agents, LangChain/Graph, RAG, CV2, Matplotlib, Seaborn

World Models & RL: DreamerV3, Dreamer 4, OpenVLA, DeepMind Control Suite, Meta-World, continual RL

Big Data & Dev: Hadoop, AWS, HTML & CSS, Ruby on Rails, Flutter, Flask, Git, Bash, Visual Studio

Simulation: Unity3D, Blender, OpenXR, OculusVR

Others: LLVMs, Adobe Suite, Office Suite, Systems Assembly

ACADEMIC SERVICES

Reviewer @ [Neural Information Processing Systems \(NeurIPS\)](#), 2025